Induwara Sirimanna

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WORK EXPERIENCES

Associate Software Engineer – Coullax **06/2025 – Present**

- Implement high-availability distributed systems using advanced data structures & algorithm design.
- Participate in peer code reviews and integration testing to ensure adherence to architectural standards.
- Identify and remediate performance bottlenecks and contribute to technical debt reduction strategies.
- Design and integrate blockchain-based solutions to enhance data integrity and transparency across distributed application layers.
- Develop and maintain multi-tenant architecture, ensuring secure tenant isolation and scalable onboarding workflows using AWS Cognito, ABAC/RBAC, and custom permission models.
- Implement AI-powered features and integrations using AWS Bedrock, including guardrail configuration for safe, responsible AI output, to extend application intelligence and automate core business processes.
- Work with AWS Aurora DB and S3 for scalable data storage and persistence across distributed application layers.

Software Engineering Internship – Coullax **12/2024 – 06/2025**

- Developed and maintained full-stack web applications using Next.js, NestJS, and the MERN stack.
- Built scalable APIs and integrated modern frontend technologies to enhance user experience.
- Worked with Docker and Kubernetes for containerization and deployment of applications, with foundational AWS experience (EC2, Amplify) from university coursework.
- Integrated AI technologies to add intelligent features to projects.
- Collaborated in agile sprints and followed best practices in code management and software development.

Web Developer (self-employed) **08/2020 - Present**

- Web Development project for Pluss Internationals PVT(LTD)
- Web Development for a real-time Flood Warning and Monitoring System (GitHub)
- Web Development project for THK Electrical Engineering PVT(LTD)

RESEARCH **06/2025 – 05/2026**

BrineX: Integrated climate-intelligent salt production ecosystem, a multimodal AI framework for sustainable yield optimization in tropic coastal environments.

The outcome: Our paper - "From Pond to Market: A Two-Stage AI Forecasting System for Salt Production Planning in Puttalam, Sri Lanka" - was accepted at ICHORA 2026, presented virtually, and is now published on IEEE Xplore. (Link for the paper)

PROJECTS

- MontiParentApp – iOS,macOS,visionOS app built for a nursery System using swift UI. (GitHub)
- BusMateSL - A Mobile App for the Bus System in Sri Lanka done using React-Native. (GitHub)
- Cleanpath - A web app for garbage collection system in Sri Lanka done using MERN stack. (GitHub)
- Pinnacle Website - A MERN stack-based game streaming and purchasing platform. (pinnacle.vercel.app)
- Cinemania - A web app for browsing movies and TV series. The project was done using JAVA. (GitHub)
- Your Way - car rental website. The project was done using PHP, HTML, SQL. (GitHub)
- My Friend App - A to-do list app with advanced features, including CRUD operations. (GitHub)
- Uniboard App - An Android app to help university students to find boarding places. (GitHub)
- Space Shooter Game App - Developed a fully functional space shooting game using Kotlin. (GitHub)

EDUCATION

Sri Lanka Institute of Information Technology **06/2020 – 06/2026**
BSc (Hons) in Information Technology specializing in Software Engineering
(WGPA – 3.09 / CGPA – 3.14)

Sandalanka Central College **01/2012 – 12/2019**

- GCE advanced level examination – B C S in the physical science field (Mathematics & ICT).
- GCE ordinary level examination – got 5A's, 3B's, 1C's (including A's for mathematics and ICT).

Certifications & Trainings

- Certificate for Participation and Presentation of Paper in ICHORA 2026 International Conference
- Free Code Camp Certification in Relational Database
- Free Code Camp Certification in Backend Development and APIs
- Free Code Camp Certification in JavaScript Algorithms and Data Structures (Beta)
- Free Code Camp Certification in Front End Development Libraries
- Aquinas Diploma in English - Foundation Level.

SKILLS

Programming Languages: JavaScript, Java, TypeScript, Python, C, C++, Kotlin, Swift
Web Development: HTML, PHP, React, React Native, Node.js, Express.js, Tailwind CSS, Next Js
Databases: SQL, MongoDB, MySQL, Database Concepts
Cloud & DevOps: AWS (Cognito, Bedrock, Aurora DB, S3, EC2, Amplify), Docker, Kubernetes, CI/CD, Permission Modeling, AI Guardrails
Software Development: OOP Concepts, Agile Methodologies, Cloud-Based Environments, ios-development
Other Skills: Technical Documentation
Version Control Systems: Git

OTHERS

Soft Skills: People Leadership, Strategic Planning, Long term Planning, Data Analysis, Training and mentoring, Budgetary, Regulatory compliance, Critical thinking

Languages: English(fluent), Sinhala(native)

Leadership:

- Group leader of projects done in the university. (2023 – 2024)
- Been Prefect at Sandalanka Central College (2018-2019)

REFERENCES

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